

Big Data Concepts & Challenges



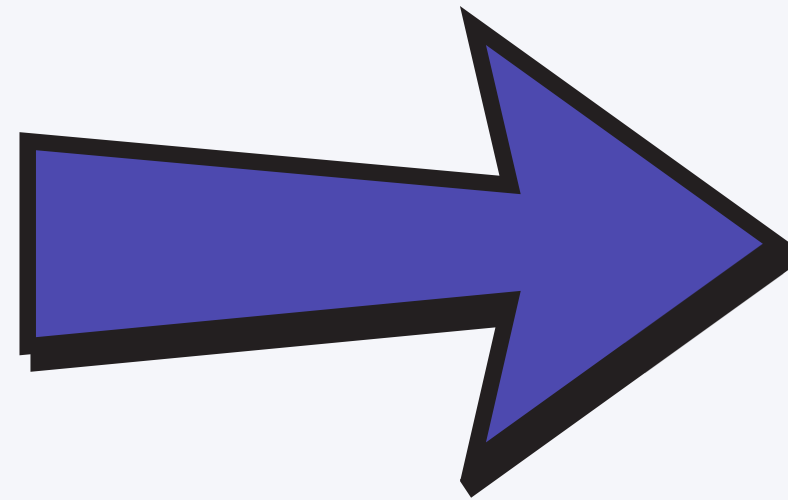
Introduction to Big Data

What is Big Data?

- Data sets so large or complex that traditional data processing applications are insufficient.

Key Insight: "Big" is relative to:

- Processing capabilities
- Storage technologies
- Analysis techniques
- Time constraints

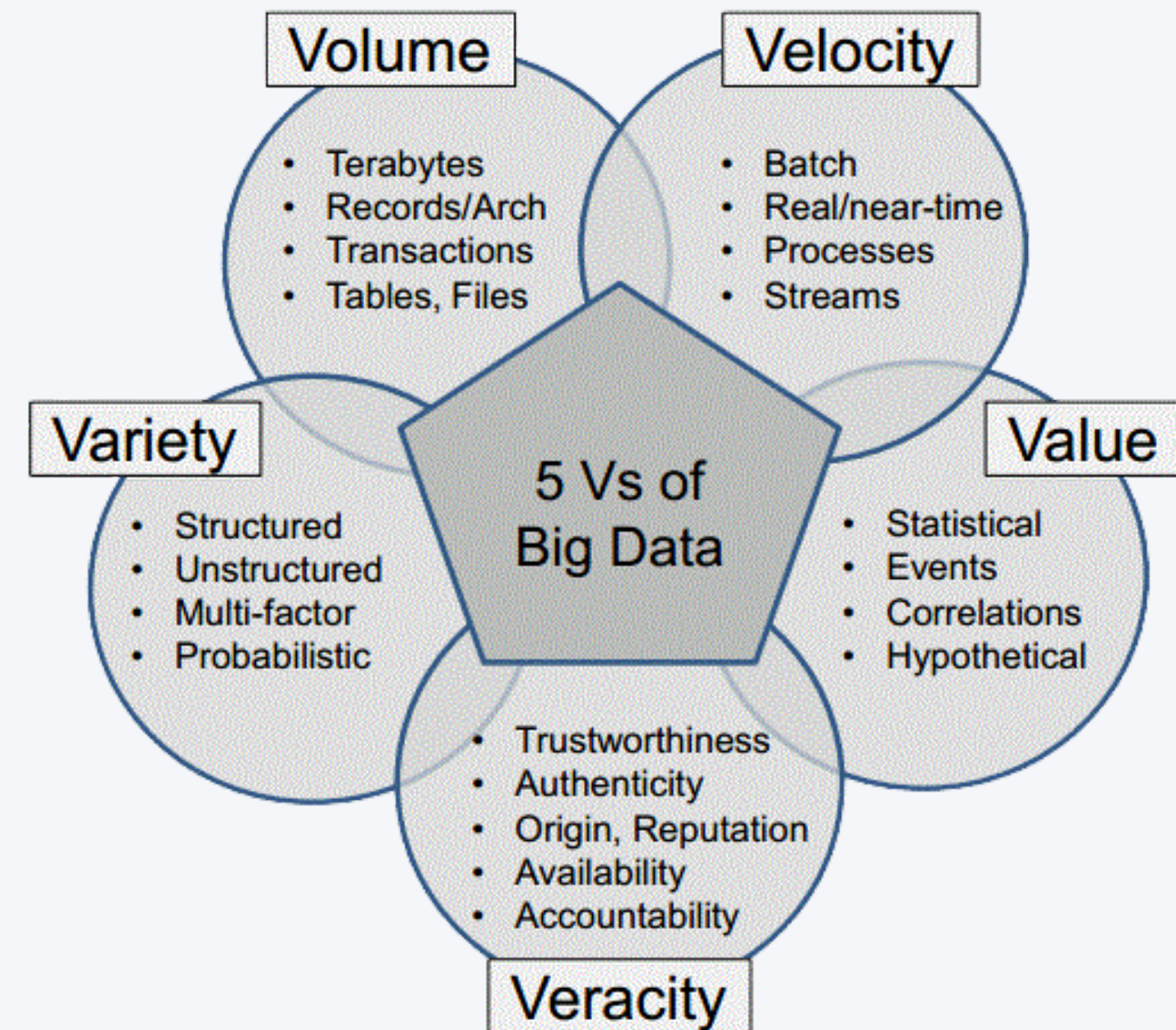


Historical Perspective:

- 1970s: **Megabytes** were "big"
- 1990s: **Gigabytes** were "big"
- 2010s: **Terabytes** became common
- 2020s: **Petabytes** and beyond

The 5 Vs of Big Data

- **Volume:** Enormous amounts of data
 - Example: YouTube users upload 500+ hours of video every minute
- **Velocity:** Speed at which data is generated and processed
 - Example: Stock market generates terabytes of data per trading day
- **Variety:** Different types and formats of data
 - Structured, semi-structured, unstructured
- **Veracity:** Trustworthiness and quality of data
 - Noise, abnormalities, biases
- **Value:** Usefulness and insights derived from data
 - Transforming data into actionable intelligence



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Volume: Scale Matters

Typical Big Data Volumes:

- Facebook: 4+ petabytes of new data per day
- Large Hadron Collider: 1 petabyte per second (filtered down)
- Internet Archive: 45+ petabytes
- Human Genome: ~200 gigabytes per sequenced genome

Storage Units:

- 1 Terabyte (TB) = 1,000 Gigabytes
- 1 Petabyte (PB) = 1,000 Terabytes
- 1 Exabyte (EB) = 1,000 Petabytes
- 1 Zettabyte (ZB) = 1,000 Exabytes

Global Data Sphere: Expected to reach 175 zettabytes by 2025

Velocity: Speed of Data

Real-time Data Generation:

- IoT sensors: Millisecond updates
- Social media: 500 million tweets per day
- Financial markets: Microsecond trading data
- Telecommunications: Call data records

Processing Requirements:

- **Batch processing:** Process data in scheduled intervals
- **Stream processing:** Continuous analysis of data in motion
- **Near real-time:** Seconds to minutes latency
- **Real-time:** Sub-second latency

Variety: Diverse Data Types

Structured Data (20% of all data):

- Relational databases
- CSV files
- Financial transactions
- Sensor readings with fixed format

Semi-structured Data:

- JSON, XML
- Email
- Web logs

Unstructured Data (80% of all data):

- Text documents
- Audio, video, images
- Social media content
- Scientific data (genomics, weather)

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Variety: Diverse Data Types

Semi-structured Data:

JSON Example

json

 Copy

```
{
  "user": {
    "id": 12345,
    "name": "Jamie Smith",
    "email": "jamie@example.com",
    "interests": ["photography", "hiking", "coding"],
    "account": {
      "type": "premium",
      "created": "2023-05-12"
    }
  },
  "posts": [
    {
      "id": 1,
      "title": "My Hiking Adventure",
      "tags": ["outdoors", "nature"]
    },
    {
      "id": 2,
      "title": "Coding Tips",
      "tags": ["programming", "python"],
      "comments": 15
    }
  ]
}
```

Veracity: Data Quality Challenges

Data Quality Issues:

- Missing values
- Duplicate records
- Inconsistent formats
- Measurement errors
- Sampling biases

Implications:

- **"Garbage in, garbage out"**
- 60% of data scientists' time spent cleaning data
- Data quality directly impacts model reliability

Approaches:

- Data validation at ingestion
- Statistical anomaly detection
- Data provenance tracking
- Automated data quality monitoring

The Breaking Points of Traditional Systems

Storage Limitations:

- Single machine storage capacity
- Backup and recovery time
- Storage access bottlenecks

Compute Limitations:

- Single machine CPU/memory capacity
- Algorithm complexity scaling
- I/O bottlenecks

Design Limitations:

- Schema rigidity
- Network limitations
- Management complexity

The Shift: **From scaling up to scaling out**

Distributed Storage Fundamentals

Core Concepts:

- Data partitioning/sharding
- Replication
- Consistency models
- Failure detection and recovery

Design Goals:

- Scalability
- Availability
- Durability
- Performance
- Consistency

Trade-offs:

- The CAP theorem
- **Consistency vs. latency**
- Storage vs. compute separation
- Performance vs. cost

The CAP Theorem

CAP = Consistency, Availability, Partition Tolerance

Consistency:

- All nodes see the same data at the same time
- Reads return most recent write

Availability:

- Every request receives a response
- No guarantee that response contains most recent write

Partition Tolerance:

- System continues to operate during network partitions

The Theorem:

- You can only guarantee 2 out of 3 properties
- In distributed systems, partition tolerance is required
- Therefore, choose between consistency and availability

Distributed Processing Concepts

Characteristics:

- Process large volumes of data
- High throughput, high latency
- Typically scheduled at intervals

Key Concepts:

- Data partitioning
- Parallel processing
- Fault tolerance
- Checkpointing

Common Use Cases:

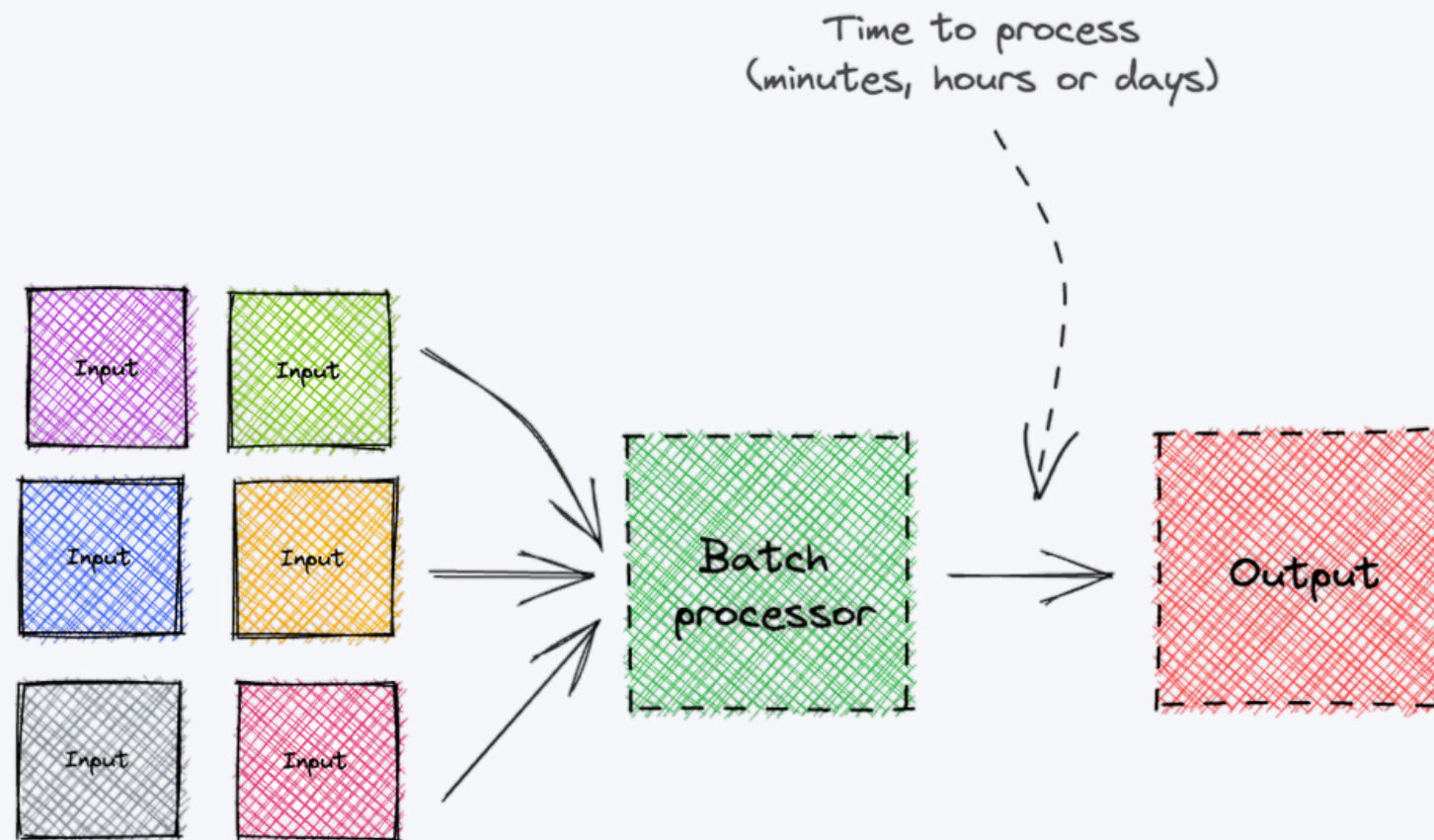
- ETL processes
- Data warehousing
- Complex analytics
- **Model training**

Programming Models:

- **MapReduce**
- Bulk Synchronous Parallel (BSP)
- Directed Acyclic Graphs (DAGs)

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Stream processing vs. Batch Processing



Batch processing

Typically used for processing large amounts of data.
Ideal for tasks that do not require real-time processing.

Stream Processing



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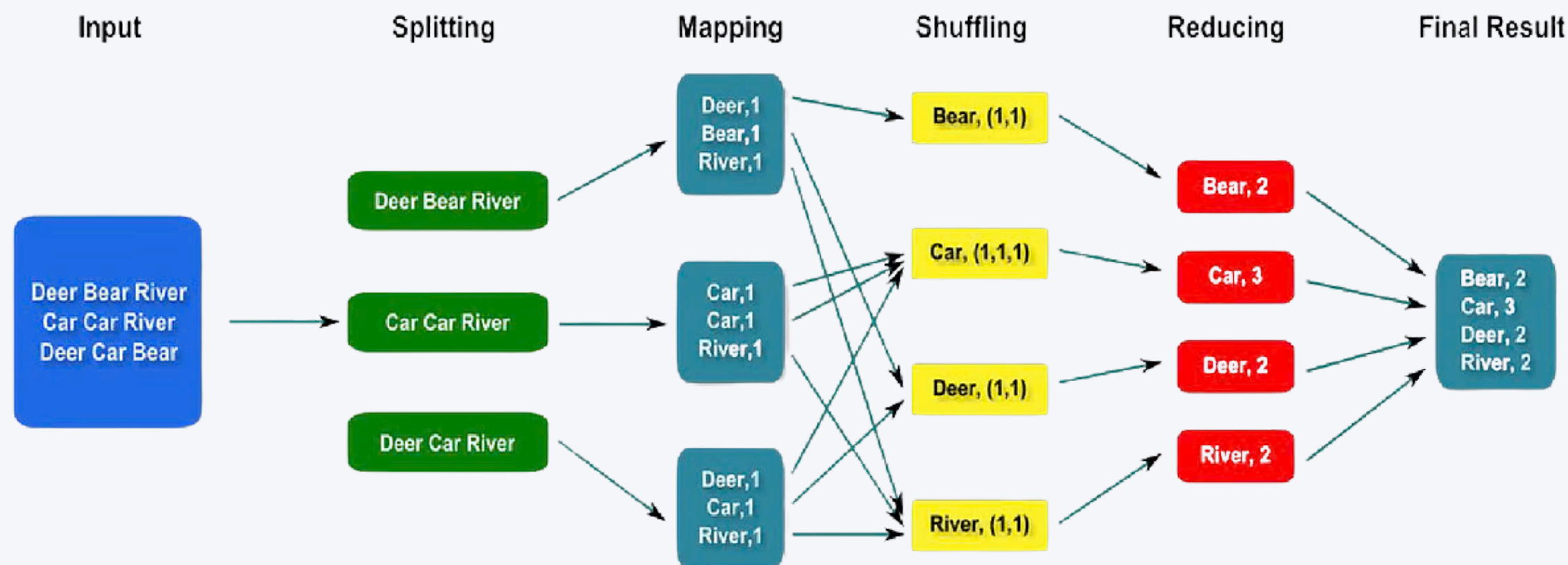
The MapReduce Paradigm

Phases:

1. **Map**: Transform and filter data (record $\rightarrow$ key-value pairs)
2. **Shuffle**: Group by key
3. **Reduce**: Aggregate results for each key

Advantages:

- Simple programming model
- Automatic parallelization
- Built-in fault tolerance
- Horizontal scalability



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The MapReduce Paradigm

Example: Customer Purchase Analysis

Imagine we have a large dataset of customer transactions that we want to analyze to find the total spending by country.

